1 **(a)** 4.07×10^{-3} **[B1]**

(b) 15×10^{10} **[M1]**

1.5×10^{11} **[A1]**

[Total Marks: 3]

2 **(a)** 4.5×10^{5} **[B1]**

(b) 0.000308 **[B1]**

(c) 8×10^{8} **[B1]**

[Total Marks: 3]

3 **(a)** 4.52×10^{5} **[B1]**

(b) 0.00607 **[B1]**

(c) 1.2×10^{10} **[B2]**

If 0 scored, M1 for 12×10^{9} **[M1]**

[Total Marks: 4]

4 **(a)** 7.14×10^{-3} **[B1]**

(b) 50200 **[B1]**

(c) 1.8×10^{10} **[B2]**

M1 for 18×10^{9} or 18000000000 seen

[Total Marks: 4]

5 **(a)** **(i)** 3.4×10^{-4} **[B1]**

(ii) 52000 **[B1]**

(b) 8×10^{8} **[B2]**

If B0, M1 for $2 \times 4 = 8$ or $10^{3} \times 10^{5} = 10^{8}$ seen.

[Total Marks: 4]

6 **(a)** 4.52×10^{-4} **[B1]**

(b) 371000 **[B1]**

(c) 8×10^{7} **[B2]**

If 0 scored, allow [M1] for 80000000 or 8×10^{k} where $k \neq 7$

[Total Marks: 4]

7 **(a)** $597 \times 10^{22} - 7.35 \times 10^{22}$ or $5.97 \times 10^{24} - 0.0735 \times 10^{24}$ **[M1]**

5.8965×10^{24} **[A1]**

(b) Sets up the correct division: $\dfrac{1.99 \times 10^{30}}{5.97 \times 10^{24}}$ **[M1]**

Evaluates correctly, e.g., $0.3333... \times 10^{6}$ **[M1]**

3.33×10^{5} **[A1]**

[Total Marks: 5]

8 **(a)** $85 \times 10^{5} - 4 \times 10^{5}$ or $8.5 \times 10^{6} - 0.4 \times 10^{6}$ **[M1]**

8.1×10^{6} **[A1]**

(b) $\dfrac{2 \times 10^{9}}{5 \times 10^{4}} = 0.4 \times 10^{5}$ or 4×10^{4} **[M1]**

$\sqrt{4 \times 10^{4}} = \sqrt{4} \times \sqrt{10^{4}}$ **[M1]**

2×10^{2} **[A1]**

[Total Marks: 5]

9 **(a)** 4.5×10^{2} **[B3]**

If 0 scored, allow [M1] for $\dfrac{3.6}{8} \times 10^{3}$ or 0.45×10^{3}

Allow a further [M1] for correctly converting to standard form, or for seeing 450

(b) 5.7×10^{8} **[B2]**

If 0 scored, allow [M1] for $(50 \times 10^{7}) + (7 \times 10^{7})$ or 570000000 or digits 57

(c) $3.2 \times 10^{2m+n+1}$ **[B2]**

If 0 scored, allow [M1] for $(4 \times 10^{2m}) \times (8 \times 10^{n})$ or $32 \times 10^{2m+n}$

[Total Marks: 7]

10 **(a)** 3×10^{8} **[B3]**

If 0 scored, M1 for $5.4 \times 10^{5} + 0.6 \times 10^{5}$ or $54 \times 10^{4} + 6 \times 10^{4}$ or 6.0×10^{5} seen **[M1]**

M1 for $\dfrac{\text{their } 6.0 \times 10^{5}}{2 \times 10^{-3}}$ **[M1]**

(b) 5×10^{4} **[B2]**

If 0 scored, M1 for $\sqrt{25 \times 10^{8}}$ or 50000 seen **[M1]**

(c) $5 \times 10^{2x-y-1}$ **[B3]**

If 0 scored, M1 for $p^{2} = 4 \times 10^{2x}$ **[M1]**

M1 for $\dfrac{4 \times 10^{2x}}{8 \times 10^{y}}$ or $0.5 \times 10^{2x-y}$ **[M1]**

[Total Marks: 8]

11 **(a)** $4 \times 10^{m+n+1}$ or $4.0 \times 10^{m+n+1}$ **[B2]**

M1 for $40 \times 10^{m+n}$ seen

(b) $3.125 \times 10^{2m-n}$ **[B3]**

M1 for $(5 \times 10^m)^2$ or 25×10^{2m} seen

M1 for $\frac{25}{8} \times 10^{2m-n}$

(c) 2 **[B3]**

M1 for $8 \times 10^n = 5.08 \times 10^m - 5 \times 10^m$

M1 for $8 \times 10^n = 0.08 \times 10^m$ or $8 \times 10^n = 8 \times 10^{m-2}$

[Total Marks: 8]

12 **(a)** 500 **[B2]**

M1 for $1.5 \times 10^8 \div 3.0 \times 10^5$

(b) 6.0×10^3 or 6×10^3 **[B2]**

M1 for $1.5 \times 10^8 \div 2.5 \times 10^4$ or 6000 seen

[Total Marks: 4]

13 **(a)** $1.23 \times 4.87 \times 10^{24}$ **[M1]**

5.99×10^{24} **[A1]**

(b) $\frac{1.99 \times 10^{30}}{4.87 \times 10^{24}}$ **[M1]**

4.09×10^5 **[A1]**

[Total Marks: 4]

14 **(a)** $\frac{1.99 \times 10^{30}}{1.90 \times 10^{27}}$ **[M1]**

$1047.36\ldots$ or $1.047\cdots \times 10^3$ **[A1]**

1.05×10^3 **[A1]**

(b) $(4.5 \times 10^{13}) \times 1000$ or $(4.5 \times 10^{13}) \times 10^3$ **[M1]**

4.5×10^{16} **[A1]**

[Total Marks: 5]

15 **(a)** 3.61×10^7 **[B2]**

If B0, M1 for $3.4 \times 10^7 + 0.21 \times 10^7$ or $34 \times 10^6 + 2.1 \times 10^6$ or 36100000 **[M1]**

(b) Converts minutes to seconds: $5 \times 60 = 300$ **[M1]**

Sets up division: $\frac{300}{4 \times 10^{-9}}$ **[M1]**

7.5×10^{10} **[A1]**

[Total Marks: 5]

16 **(a)** 6.04×10^{24} **[B2]**

If 0 scored, allow [M1] for 6.0435×10^{24} or 604.35×10^{22} or digits 604

(b) 81 **[B2]**

If 0 scored, allow [M1] for $\dfrac{5.97 \times 10^{24}}{7.35 \times 10^{22}}$ or 81.2...

(c) 1.79 **[B2]**

If 0 scored, allow [M1] for $300 \times 5.97 \times 10^{24}$ or 1.791×10^{27} or digits 179

[Total Marks: 6]

17 **(a)** 5.8965×10^{24} **[B2]**

If 0 scored, M1 for $5.97 \times 10^{24} - 0.0735 \times 10^{24}$ or $597 \times 10^{22} - 7.35 \times 10^{22}$ seen **[M1]**

(b) 5×10^2 or 5.00×10^2 **[B2]**

If 0 scored, M1 for $\dfrac{1.50 \times 10^8}{3.00 \times 10^5}$ or 500 **[M1]**

(c) 5.10×10^{14} **[B3]**

If 0 scored, M2 for $4 \times \pi \times (6.37 \times 10^6)^2$ **[M2]**

If 0 or M2 not scored, M1 for $(6.37 \times 10^6)^2$ or $4.057... \times 10^{13}$ **[M1]**

[Total Marks: 7]

18 **(a)** $\pi \times (4.5 \times 10^2)^2 \times (1.2 \times 10^3)$ **[M1]**

$7.63407 \cdots \times 10^8$ **[A1]**

7.634×10^8 **[A1]**

(b) Volume in $m^3 = \dfrac{7.634 \times 10^8}{10^6} = 763.4\ m^3$ (or equivalent rate conversion $1.5 \times 10^2\ cm^3/\text{s}$) **[M1]**

Time in seconds $= \dfrac{763.4}{1.5 \times 10^{-4}} = 5.089 \cdots \times 10^6$ **[M1]**

Time in hours $= \dfrac{5.089 \cdots \times 10^6}{3600} = 1413.7 \ldots$ **[M1]**

1.41×10^3 **[A1]**

[Total Marks: 7]